Review plan · for the teacher

Data in your school: unit check

Data engineering · Level 1 · Unit check

Time	Materials
40 min	Board, chalk, paper. No devices needed.

Learning goals

- Data all around the school
- Ask a question, then collect
- Charts and summaries that tell the truth
- Pupils' data needs care

Activity: Team quiz

1. Teams (2 min): Make teams of four and give each team a sheet of paper for its answers.
2. Quiz (20 min): Read each question aloud from the quiz sheet; each team agrees on one answer and writes it down.
3. Explain (10 min): Reveal each answer and ask one team to explain why it's right, using an example from the unit.
4. Wrap-up (8 min): Write the question most teams got wrong on the board, and work through it together.

Discussion

- Which idea from this unit will you use first?
- Which sheet in our school would you tidy first?
- What will you change about how you share pupils' data?

Exit ticket

- One habit I will use this week, and where I will use it.

Try it in class tomorrow
Pick one sheet your school keeps, make a tidy copy with pupil numbers instead of names, and draw one honest chart from it with your class.

Quiz sheet · for each team

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Team _____ Date _____

1 **In a tidy table of pupils, what is one column?**

◯ One pupil ◯ One fact about every pupil, such as their class ◯ The total of the marks

2 **A Total row inside the data can make a computer's average wrong.**

◯ True ◯ False

3 **Which of these make a good collection sheet? Choose all that apply.**

☐ Fixed codes such as P, A and L ☐ Dates written one way, such as 2026-03-02
☐ The unit in each column heading ☐ Columns for everything, just in case
☐ Dates written however each person likes

4 **Match each word to what it means.**

CSV	• • • •	The middle value once the values are sorted
Tally marks	• • • •	A table as plain text, with commas between the values
Median	• • • •	Facts such as health or religion that need the most care
Sensitive data	• • • •	Lines for counting, in groups of five

5 **Your survey's minutes are 12, 15, 9, 14 and 400 (a typo for 40). Which summary is most affected by the typo?**

◯ The count ◯ The median ◯ The mean

6 **Put the steps in order: from a question to a chart.**

☐ Collect with tally marks
☐ Write the question
☐ Put the counts in a tidy table
☐ Draw the chart, with bars from zero
☐ Plan the columns, units and codes

7 **Saving the same data as JSON instead of CSV can make it more than twice as big.**

◯ True ◯ False

8 **Which of these is fine to post in the parents' WhatsApp group?**

◯ The class average for the test ◯ A list of the pupils who failed ◯ Each pupil's test mark, by name

9 **Which habits keep pupils' data safe? Choose all that apply.**

☐ Keep everything forever, just in case ☐ Keep it locked with a PIN or a key
☐ Delete it when it's no longer needed ☐ Share it with anyone who asks politely ☐ Collect only what you need

10 **Complete the rules of a tidy table.**

One _____ per thing, one _____ per fact, one value per _____.

Word bank: chart · row · total · cell · column

Answers · for the teacher

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1 **One fact about every pupil, such as their class**

Rows are things (here, pupils) and columns are facts about them.

2 **True**

The computer counts the Total row as one more pupil with a huge mark, so the average comes out far too high.

3 **The unit in each column heading; Fixed codes such as P, A and L; Dates written one way, such as 2026-03-02**

Units in headings, fixed codes and one date format make everyone write the same way.

4 **CSV — A table as plain text, with commas between the values**
Tally marks — Lines for counting, in groups of five
Median — The middle value once the values are sorted
Sensitive data — Facts such as health or religion that need the most care

CSV stores tables as text, tally marks count in fives, the median is the middle of sorted values, and sensitive data is personal data that can do special harm if shared.

5 **The mean**

The mean adds every value, so 400 drags it far up.

6 **Top to bottom: 3, 1, 4, 5, 2**

The question decides what to collect and how the sheet looks; then you collect, organise the counts in a tidy table, and only then draw the chart.

7 **True**

JSON repeats every column name in every record, so the same facts take more than twice the space: more time and money to send on a slow connection.

8 **The class average for the test**

The class average tells parents how the class did without pointing at any child.

9 **Collect only what you need; Keep it locked with a PIN or a key; Delete it when it's no longer needed**

Collecting less, locking it and deleting it on time all protect pupils.

10 **One row per thing, one column per fact, one value per cell.**

Rows are things, columns are facts, and each cell holds a single value.